Stefan Free Boundary Problem

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0.1 The Stefan Problem

0.1.1 Description and Connection with Cold Spray

The Stefan problem describes heat transfer in a material containing a moving phase boundary, where the interface between two phases evolves over time as latent heat is absorbed or released. A classic example is the melting or freezing of ice submerged in a body of water, where the solid–liquid interface moves in response to heat conduction.

A particularly important engineering application of Stefan-type problems is welding. In conventional welding processes, metal at the joint interface is heated above its melting temperature, with or without the addition of a filler material, before solidifying to form a strong metallurgical bond. Cold spray deposition shares several characteristics with explosive welding, although the underlying mechanisms differ. During cold spray, metallic particles impact the substrate at supersonic velocities, producing severe plastic deformation and localized adiabatic heating. These conditions promote adiabatic shear instability and oxide-layer disruption, enabling solid-state metallurgical bonding between the particles and the substrate without bulk melting.[1]

0.1.2 The Mathematical Problem

We will consider the 2 phase Stefan problem in 2 dimensions. [2]

We denote with Ω_i as the region for each phase i, $T_i(\vec{x}, t)$ as the temperature in the respective region, $\partial\Omega_i$ as the boundaries, α_i as the thermal diffusivity, and Q_i as sources of heat.

Within each region, it will satisfy the inhomogeneous heat diffusion equation with their respective thermal diffusivity and Robin boundary conditions. One specific condition is that the temperature at the free boundary be equivalent to the critical phase transition temperature.

$$\frac{\partial T_i}{\partial t} = \nabla \cdot (\alpha_i \nabla T_i) + Q_i \text{ ; for } \vec{x} \in \Omega_i \tag{1}$$

$$a(x)T_i(x) + b(x)\partial_n T_i(x) = g(x) \text{ ; for } \vec{x} \in \partial\Omega_i \tag{2}$$

The interface transitions between each phases i and j with a free surface velocity $\vec{V}$ that satisfies the following differential equation.

$$\vec{V_i}(x,t) = \frac{\mathrm{d}x}{\mathrm{d}t} \text{ ; for } \vec{x} \in \partial\Omega_i \tag{3}$$

$$L_{ij}\vec{V_i} = \alpha_i \partial_{n_i} T_i - \alpha_j \partial_{n_j} T_j \text{ ; for } \vec{x} \in \partial\Omega_i \cap \partial\Omega_j \tag{4}$$

With L_{ij} being the latent heat of phase transition between phases i and j, and ∂_{n_i} being the limit of the outward normal derivative of Ω_i.

0.2 FreeFEM Implementation

0.2.1 The Chosen Toy Problem

We choose the classic problem of ice melting and solidifying. CGS units are used here.

The necessary constants are stated in the table below.

Name	Notation in code	Numerical value
Thermal diffusivity of ice	k_{ice}	1.335E-2 cm^2/s
Thermal diffusivity of water	k_{water}	1.4E-3 cm^2/s
Latent heat of melting	L	334E-2 cm^2/s^2

Where we have chosen to scale the latent heat of melting by 1E-11 to amplify the effects of melting for visual purposes.

We consider the problem in 1cm by 1cm square bisected by the given parametric function.

$$x(t) = 0.3 + 0.4\frac{\frac{1}{1+\exp(-k(t-0.5))} - \frac{1}{1+\exp(0.5k)}}{\frac{1}{1+\exp(-0.5k)} - \frac{1}{1+\exp(0.5k)}} \tag{5}$$

$$y(t) = 1 - t \tag{6}$$

where the parameter k determines the smoothness of the jump. The region on the left will be ice, and the region on the right will be water. See Fig 1

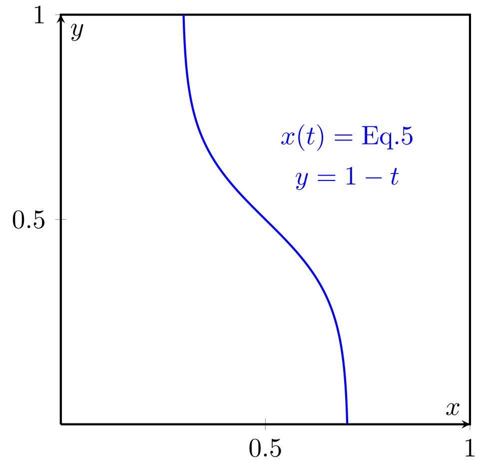


Figure 1: The curve bisecting the 2 phases. The region on the left will be ice, and the region on the right will be water.

We choose the below boundary conditions.

$$\frac{\partial T_{ice}}{\partial n} = 0 \text{ ; for (x,y) on the left, upper, and lower edges.} \tag{7}$$

$$T_{ice} = 0 \text{ ; for (x,y) on the free surface.} \tag{8}$$

$$\frac{\partial T_{water}}{\partial n} = 0 \text{ ; for (x,y) on the upper and lower edges.} \tag{9}$$

$$T_{water} = 0 \text{ ; for (x,y) on the free surface.} \tag{10}$$

$$T_{water} = 5 \text{ ; for (x,y) on the right edge.} \tag{11}$$

The initial temperatures will be -1 for ice, and 1 for water everywhere. Finally, with no external heat source $- Q = 0$.

0.2.2 The Numerical Implementation

We build the meshes separately to obtain the normal vectors at the boundaries, and assign the appropriate regions as ice and water by use of an indicator function "regIce".

```
// Left region (ice)
mesh ThL = buildmesh(S1(n) + S2(n) + S3(n) + S4(n) + D1(n));

// Isolate left region (ice): region containing point (0.1,
    0.5)
int regIce = ThL(0.1, 0.5).region;
mesh ThIce   = trunc(ThL, region == regIce);
mesh ThWater = trunc(ThL, region != regIce);
```

```
mesh IceStart= ThIce;
```

We then build the piecewise quadratic finite element space.

```
fespace ThI(ThIce, P2);
ThI dxI,dyI, i1, i2; // Mesh moving fespace
ThI iu, iuold, iv; // temperature fespace

fespace ThW(ThWater, P2);
ThW dxW,dyW, w1, w2;
ThW wu, wuold, wv;
```

Where we keep the old temperature to use the Crank-Nicholson solver.

We then solve the homogeneous heat diffusion problem with Crank-Nicholson.

```
    real theta = 0.5; // Crank-Nicholson

problem HeatICN(iu, iv)
    = int2d(ThIce)(
          iu*iv/dt
        + theta * kice * (
              dx(iu)*dx(iv)
            + dy(iu)*dy(iv)
        )
    )
    - int2d(ThIce)(
          iuold*iv/dt
        - (1-theta) * kice *(
              dx(iuold)*dx(iv)
            + dy(iuold)*dy(iv)
        )
    )
    + on(diag,  iu=0)
    ;
problem HeatWCN(wu, wv)
    = int2d(ThWater)(
          wu*wv/dt
        + theta * kwater * (
              dx(wu)*dx(wv)
            + dy(wu)*dy(wv)
        )
    )
    - int2d(ThWater)(
          wuold*wv/dt
        - (1-theta) * kwater *(
              dx(wuold)*dx(wv)
            + dy(wuold)*dy(wv)
        )
    )
    + on(right, wu=5)
    + on(diag,  wu=0)
```

```
    ;
```

We also set up a Laplace problem for moving the boundary, which will be elaborated upon in the next subsubsection.

```
    problem MoveIX(dxI, i1) =
    int2d(ThIce)(dx(dxI)*dx(i1) + dy(dxI)*dy(i1))
  + on(left, dxI = 0)
  + on(diag,  dxI = dt/L*(kice*dx(iu)*N.x-kwater*dx(wu)*N.x)
     )
  ;

problem MoveIY(dyI, i2) =
    int2d(ThIce)(dx(dyI)*dx(i2) + dy(dyI)*dy(i2))
  + on(diag,  dyI = dt/L*(kice*dy(iu)*N.y-kwater*dy(wu)*N.y)
     )
  + on(top, dyI = 0)
  + on(bottom, dyI = 0)
  ;

problem MoveWX(dxW, w1) =
    int2d(ThWater)(dx(dxW)*dx(w1) + dy(dxW)*dy(w1))
  + on(right, dxW = 0)
  + on(diag,  dxW = -dxI)
  ;

problem MoveWY(dyW, w2) =
    int2d(ThWater)(dx(dyW)*dx(w2) + dy(dyW)*dy(w2))
  + on(diag,  dyW = -dyI)
  + on(top, dyW = 0)
  + on(bottom, dyW = 0)
  ;
```

0.2.3 Challenge with Moving the Boundary

We quickly ran into a challenge with moving the boundary.

The numerical formulation for the boundary move increment Δx is given by the following equation.

$$\Delta x_i = \Delta t \cdot \vec{V} = \frac{\Delta t}{L}\left(\alpha_i \partial_{n_i} T_i - \alpha_j \partial_{n_j} T_j\right) \tag{12}$$

The first numerical formulation on incrementing the boundary only moved the finite element points on the boundary. This causes some of the finite elements to flip and FreeFEM terminates with an error of negative area.

To resolve the error, one has a few choices. The simplest way forward is to adjust the time step such that the boundary does not move enough to flip the finite elements, and then remesh to eliminate bad mesh geometry at the boundary. However, this approach severely restricts the time discretization stepsize by the size of the finite elements, and is difficult to predict the appropriate step size.

Naturally then, the choice will be to move all points, and remesh whenever necessary. The problem now is to interpolate the increment everywhere within the domain given the boundary increment. Thus, we choose to solve the following Laplace problem.

$$\nabla^2 \vec{X}_i = 0 \quad \text{in } \Omega_i \tag{13}$$

$$\vec{X}_i = \Delta x_i \quad \text{on } \partial\Omega_i \cap \partial\Omega_j \tag{14}$$

Where $\vec{X}_i$ will be the mesh increment everywhere in Ω_i. Given all the regularization theorems regarding the Laplace operator, one can guess that this allows a much wider range of time step sizes to be used. We present no analysis on the validity of this guess here.

0.2.4 Results

The results can be visualized as a time series in Paraview, or any .vtu viewer.

The ice region grows in size as it solidifies due to a higher thermal diffusivity, as it loses heat energy into the water region. It then stops growing and begins to melt as the water which is maintained at 5 degree celsius on the right bound inputs heat into the ice region.

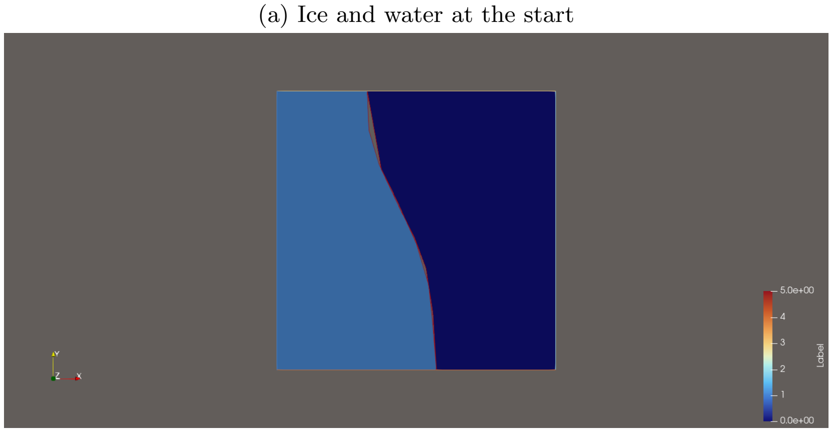
In the end, we get a net loss of ice region.

References

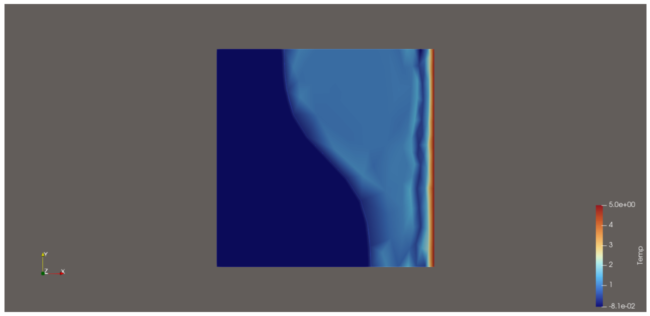
[1] H. Assadi, H. Kreye, F. Gärtner, and T. Klassen. Cold spraying – a materials perspective. *Acta Materialia*, 116:382–407, 2016.

[2] Peter Howell. Free boundary problems. Lecture notes, InFoMM Specialist Course: Continuum Models in Industry, University of Oxford, 2015. Accessed: 2026-08-05.

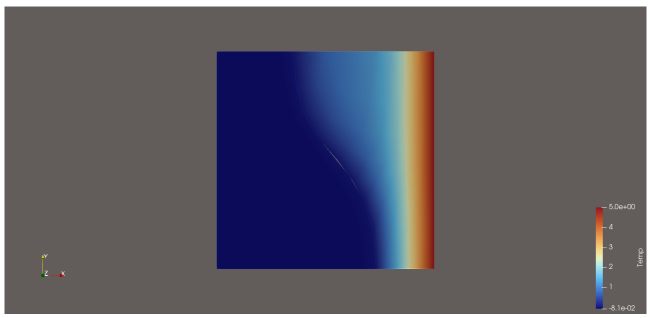
(a) Ice and water at the start



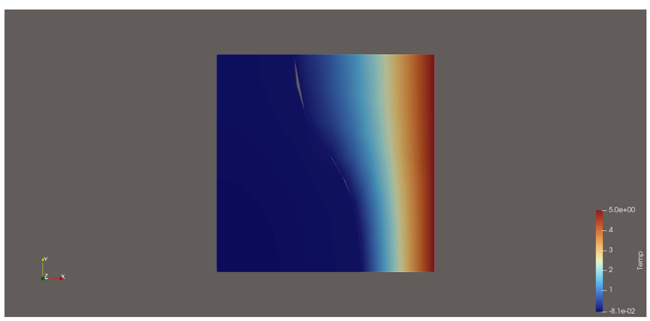
(b) Ice and water at the end



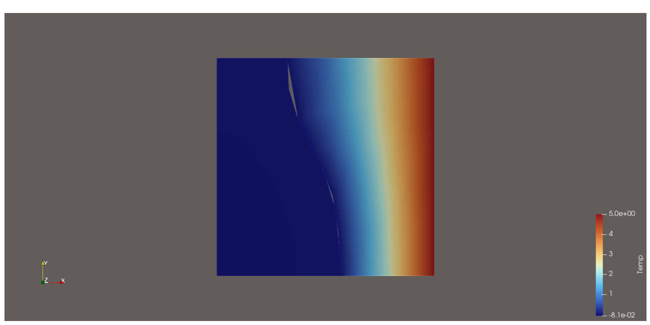
(a) t=0s



(b) t=7.5s



(c) t=25s



(d) t=50s